

Safety Helmet Detection Using YOLOv8

This project implements a YOLOv8n object detection system for workplace safety images. The system detects person, helmet, and head.

The Hard Hat Detection dataset was converted from Pascal VOC XML format into YOLO TXT format.

A total of 150 images were selected: 105 training images, 30 validation images, and 15 test images.

The model was trained for 30 epochs with image size 640 and batch size 4 on CPU.

Validation results were precision 0.838, recall 0.446, mAP@0.5 0.511, and mAP@0.5:0.95 0.283.

Test results were precision 0.820, recall 0.418, mAP@0.5 0.467, and mAP@0.5:0.95 0.263.

The helmet class performed best with test mAP@0.5 of 0.779. Person detection was moderate with mAP@0.5 of 0.599.

The head class performed poorly because it was underrepresented in the dataset.

The main errors were missed persons, loose boxes, overlapping detections, missed small helmets, and weak head predictions.

The original annotations were converted and visually reviewed. The head class is not automatically equivalent to no_helmet.